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Introduction

Embarking on a journey in the dynamic realm of technology, I am a dedicated and forward-thinking individual with a passion for innovation. With a foundation built on four years of hands-on experience in the Immersive Technology landscape, I have actively chosen the path of continuous learning and embraced challenges . As a naturally curious mind, I find joy in unraveling complexities and encouraging others to express their thoughts openly. My enthusiasm lies in exploring the ever-changing digital landscape and harnessing its potential to create impactful and accessible solutions.

1. Mahavitharan

Problem :
Developing safety modules for electrical workers to mitigate injuries among linemen posed a significant challenge. The objective was to create a realistic training experience without the risk of working with actual high-voltage equipment



Mahavitharan

My contribution & Approach:

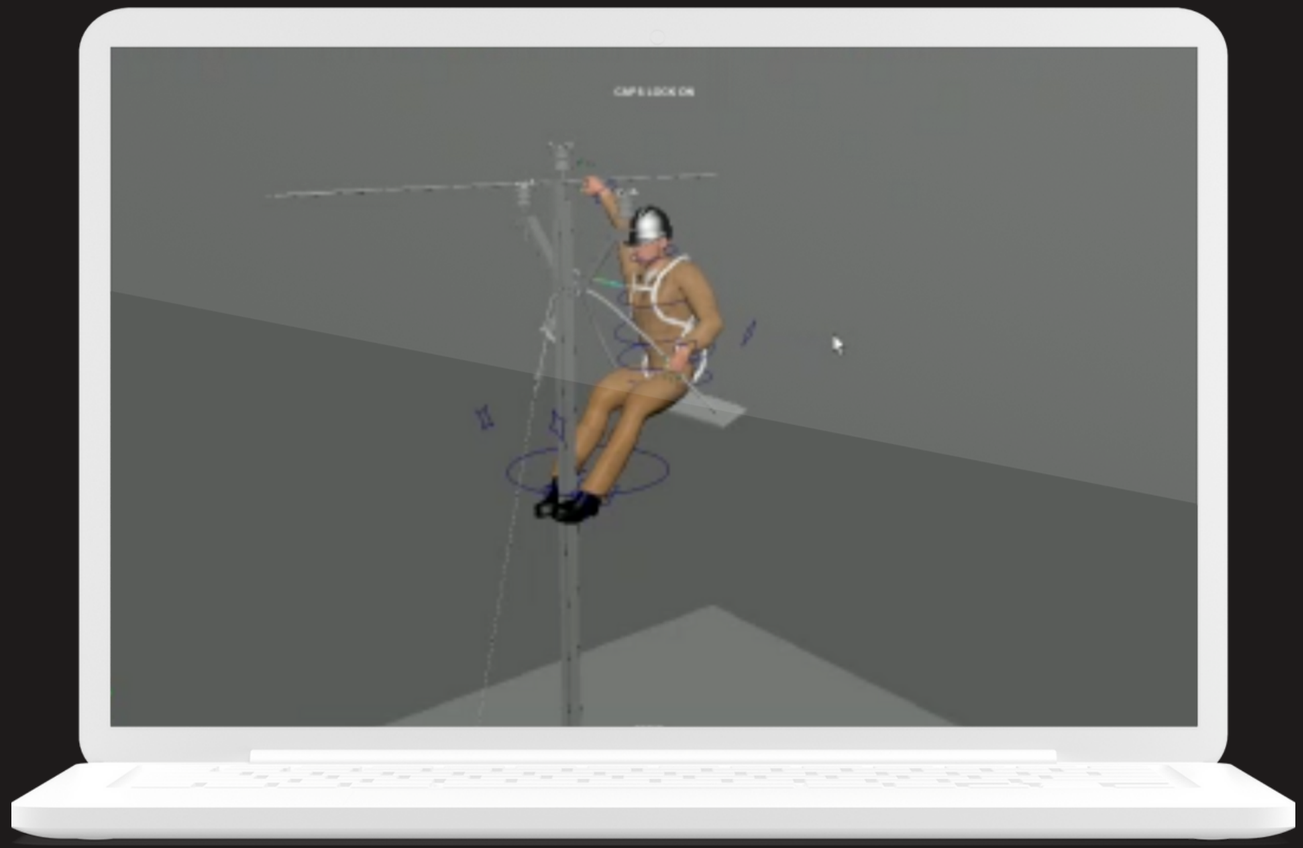
I led the development of the entire project, bringing together geographically dispersed cross-functional teams. Spearheading the software development efforts, We utilized VR headsets, motion capture suits with haptic devices, and a 3D scene mirroring the substation's layout. This enabled linemen to practice operations such as transformer coil replacement and jumper restringing.



Mahavitharan

Impact:

Our innovative approach resulted in an effective training tool that reduced the risk of injuries among linemen by 41%. By providing a safe environment for practice, we ensured linemen were well-prepared before working with high-voltage equipment. The detailed industrial plant design and segmented work approach minimized rework and led to project success.



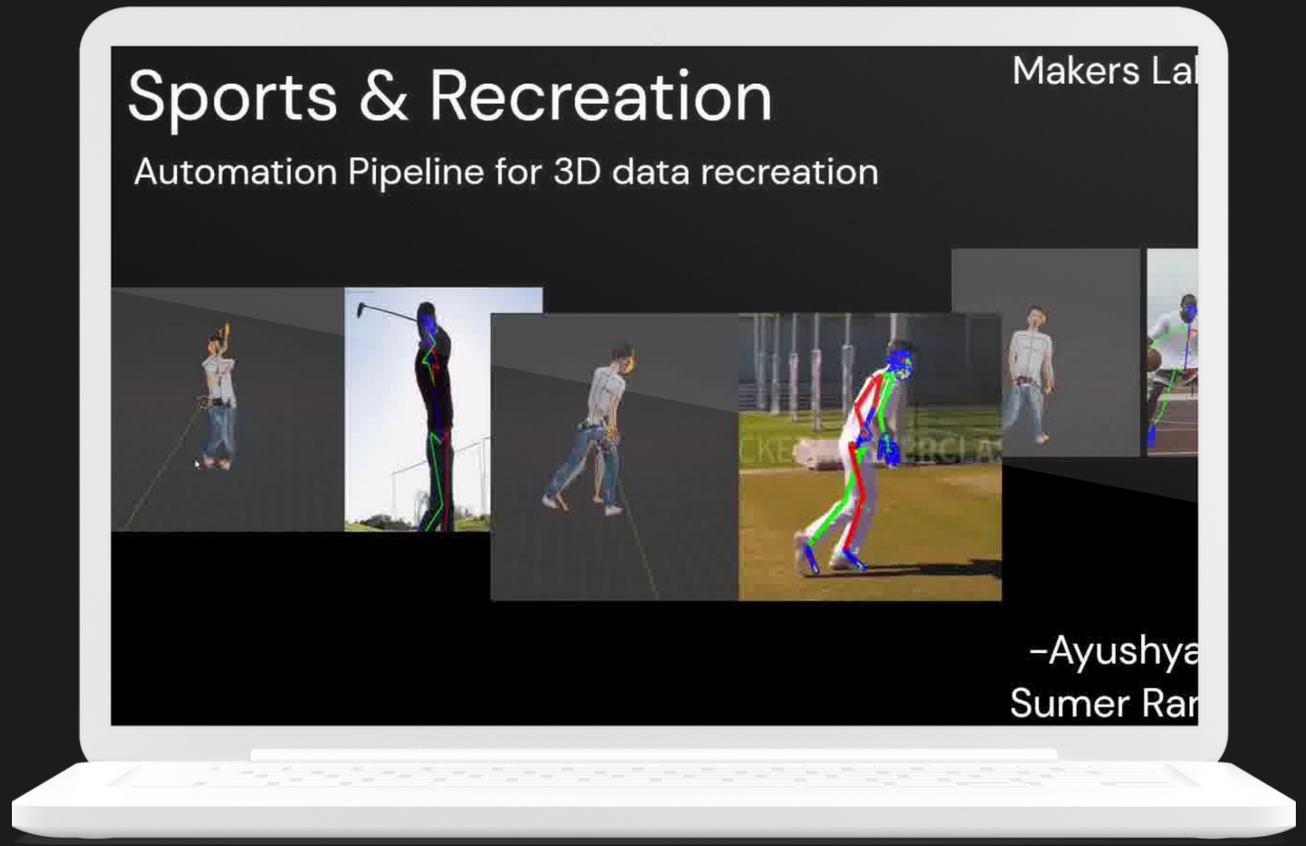
Mahavitharan

Learning:
Leading the development of Mahavitharan enhanced my leadership skills, customer interaction, and ability to bring innovation to a wider audience. The project instilled resilience and perseverance, and ensuring success involved strategic planning and iterative development of a hardware product. This experience significantly contributed to my growth and proficiency in immersive technology.



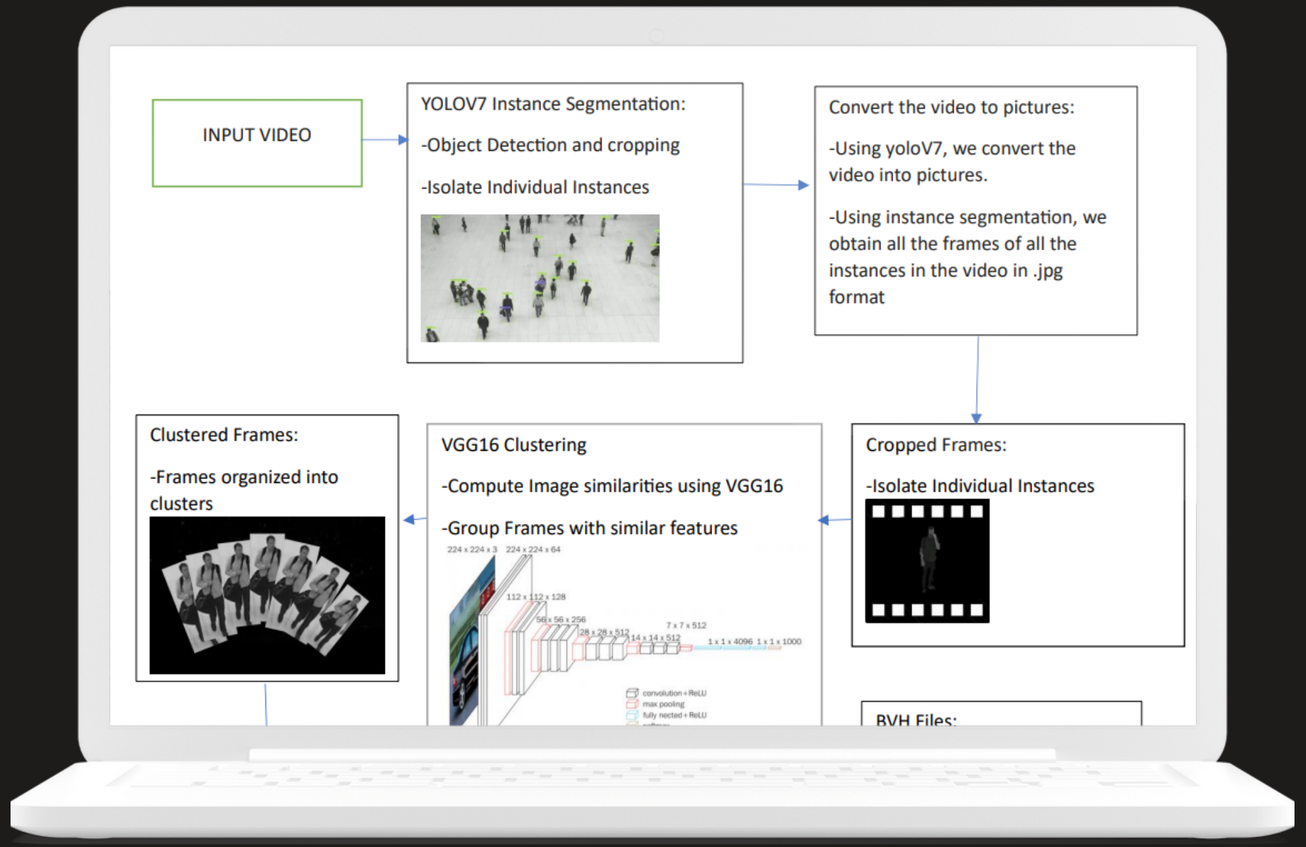
2.Root based 3D reconstruction of multiple human pose

Problem:
The challenge was to recreate dynamic scenes from videos containing human subjects, offering transformative applications in sports viewing, analytics and investigations. The field of scene recreation can revolutionize coaching strategies by providing detailed insights into player movements and team dynamics.



Root based 3D reconstruction of multiple human pose

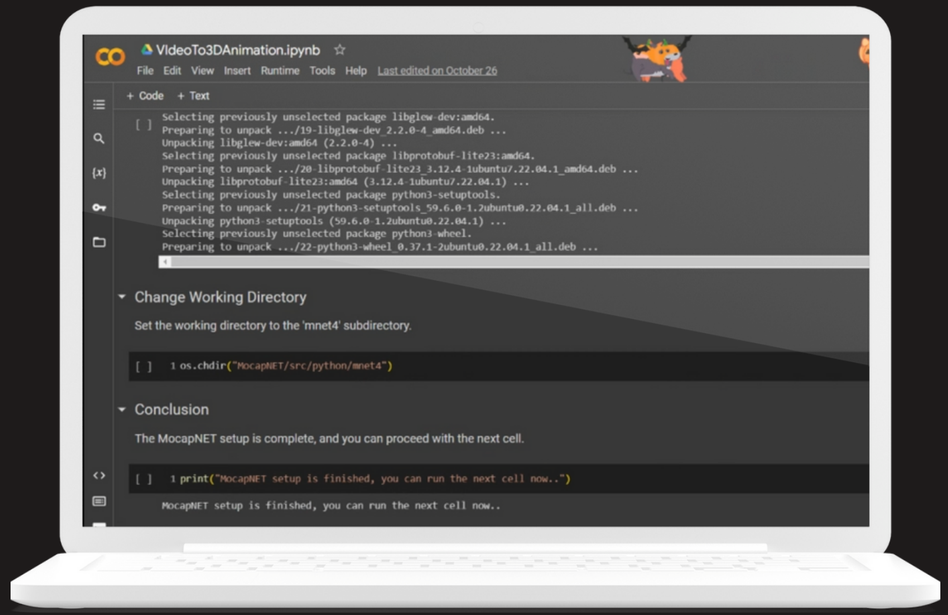
Approach:
We proposed a novel pipeline that involved converting 2D images representing the poses to 3D humanoids in a virtual environment. We utilized techniques such as instance segmentation, clustering, convolutional neural networks, and MocapNet for accurate pose estimation and representation. We also integrated the generated 3D poses into Blender for visualization.



Root based 3D reconstruction of multiple human pose

My Contribution:

In this research I proposed and implemented the technical pipeline. My contributions included designing and leading the research, where I employed techniques such as instance segmentation, clustering, convolutional neural networks using MocapNet for accurate pose estimation and representation. As a first-time participant in the research paper lifecycle, I took on the responsibility of writing the paper, ensuring clarity in presenting our methodologies and results.

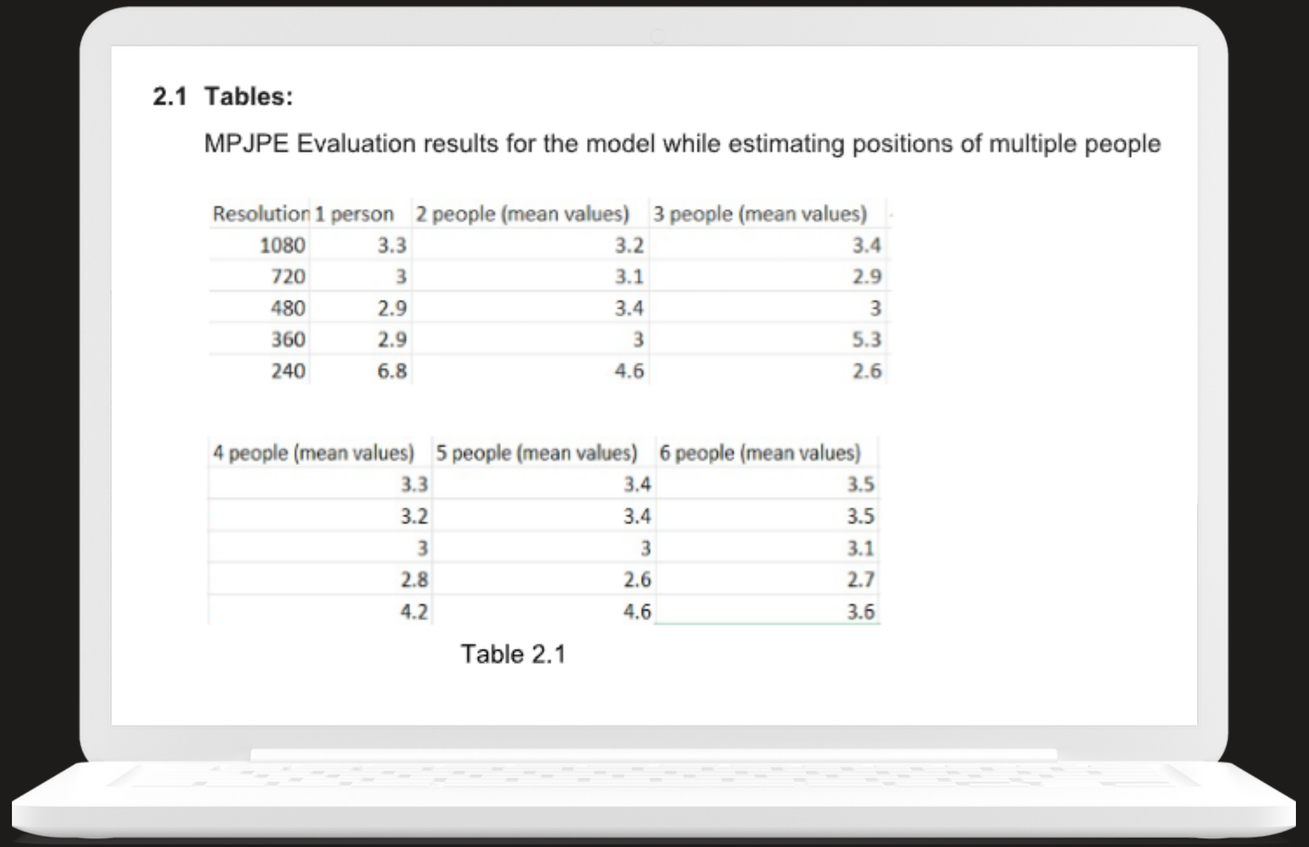


Additionally, I supervised the benchmarking and data collection efforts for the project, collaborating with an intern to collect and analyze the necessary data. The successful outcome of this research demonstrated the potential of our proposed techniques for advancing 3D pose estimation and also marked a significant milestone in my journey of contributing to cutting-edge advancements in human-computer interaction.

Root based 3D reconstruction of multiple human pose

Impact:

The developed system successfully reconstructed 3D human poses in a virtual environment, allowing for dynamic poses and movement around the root base. The use of BioVision Hierarchy (BVH) files facilitated compatibility with various 3D graphics engines. The system showed commendable accuracy, as evaluated through mean per joint position error (MPJPE) on the H36M dataset. The results demonstrated the potential of our proposed techniques for advancing the field of 3D pose estimation and representation in virtual environments.



2.1 Tables:

MPJPE Evaluation results for the model while estimating positions of multiple people

Resolution	1 person	2 people (mean values)	3 people (mean values)
1080	3.3	3.2	3.4
720	3	3.1	2.9
480	2.9	3.4	3
360	2.9	3	5.3
240	6.8	4.6	2.6

4 people (mean values)	5 people (mean values)	6 people (mean values)
3.3	3.4	3.5
3.2	3.4	3.5
3	3	3.1
2.8	2.6	2.7
4.2	4.6	3.6

Table 2.1

Root based 3D reconstruction of multiple human pose

Learning:

Leading the Research of a root-based 3D reconstruction system for multiple human poses was an enriching experience that deepened my understanding of various aspects of the research process sharpening my critical thinking and problem-solving abilities. Beyond the technical intricacies, I honed crucial skills in literature review, experimental design, and data analysis. Additionally, managing the benchmarking and data collection aspects of the project enhanced my organizational and project management skills.



This experience not only contributed to the advancement of human-computer interaction but also equipped me with a broader set of research skills, fostering a continuous commitment to intellectual growth and exploration in emerging technologies.

3.Manav

Problem:
Metahumans often face limitations as they are reduced to emotionless, speechless, and brainless digital replicas. This challenge hampers their ability to authentically engage with humans in various applications. Overcoming these limitations requires advancements in natural language processing, emotional intelligence modeling, and cognitive computing. Finding solutions to enhance metahuman capabilities is crucial for creating more realistic and effective interactions in the digital realm.



Manav

Approach:

The Metahuman Lip Sync SDK employs a sophisticated approach to achieve accurate and realistic lip synchronization in digital humans. Beginning with advanced speech processing techniques, the framework utilizes automatic speech recognition (ASR) to transcribe spoken words into text. Subsequently, a phoneme analysis algorithm is applied to generate precise phonetic representations. Leveraging these representations, the SDK dynamically manipulates lip movement through blend shapes, ensuring precise synchronization with the corresponding speech.

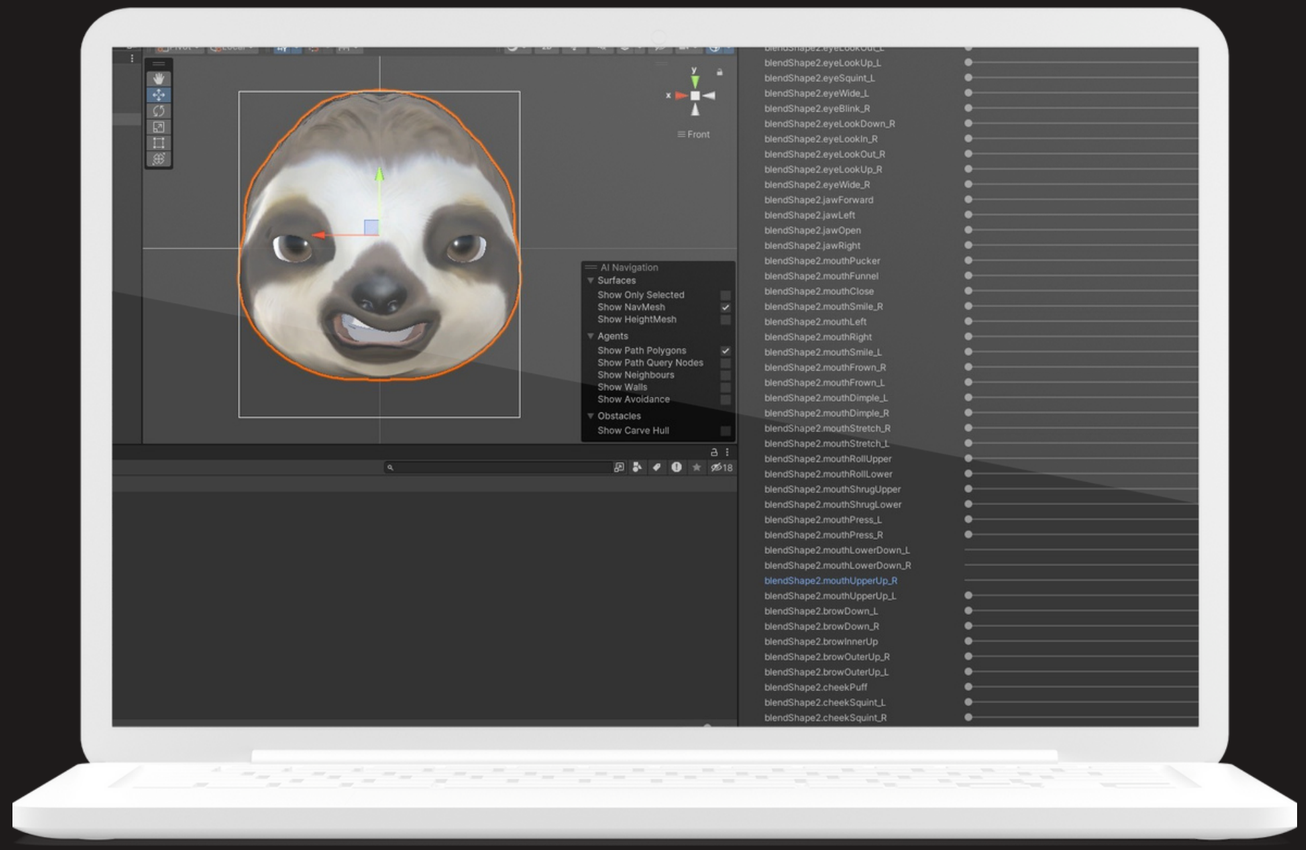


The approach involves real-time processing capabilities, guaranteeing seamless coordination of lip movements for an immersive metahuman experience. By converting words into phonemes and translating them into facial actions, the SDK achieves a high level of accuracy and realism, creating digital humans that authentically recreate speech and facial expressions.

Manav

My Contribution:

In my role at Manav, I shaped the technical pipeline, streamlining the entire communication process. My responsibilities extended to leading the team through the project's implementation phase, ensuring the seamless execution of the system design. I actively contributed to the development of the schema, focusing on load management and problem-solving to ensure the project's successful transition to a live, full-scale application.



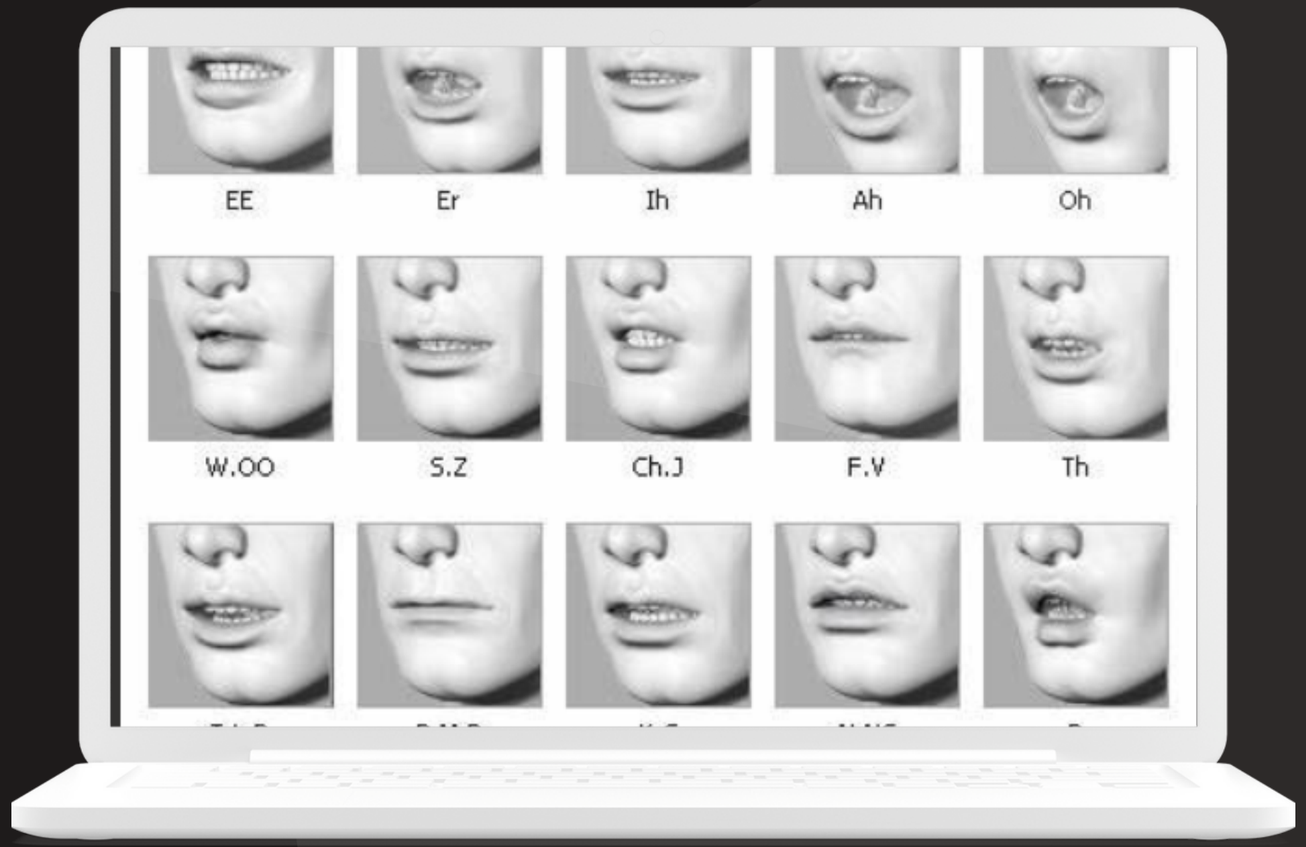
Manav

Impact:
As a result of our approach, we developed an emotional intelligence system that detects emotions and applies blend shapes corresponding to those emotions. This significantly improved the conversion process, reducing the response time from up to 60 seconds to less than a second.



Manav

Learnings:
I gained unique insights as I contributed to scaling a fully-fledged product from its inception which enabled me to foresee potential challenges and points of failure, promoting a proactive problem-solving approach. Throughout this journey, I acquired practical knowledge in technical skills, teamwork, and effective troubleshooting. This comprehensive experience significantly deepened my understanding of project dynamics and bolstered my ability to navigate the complexities of application production.



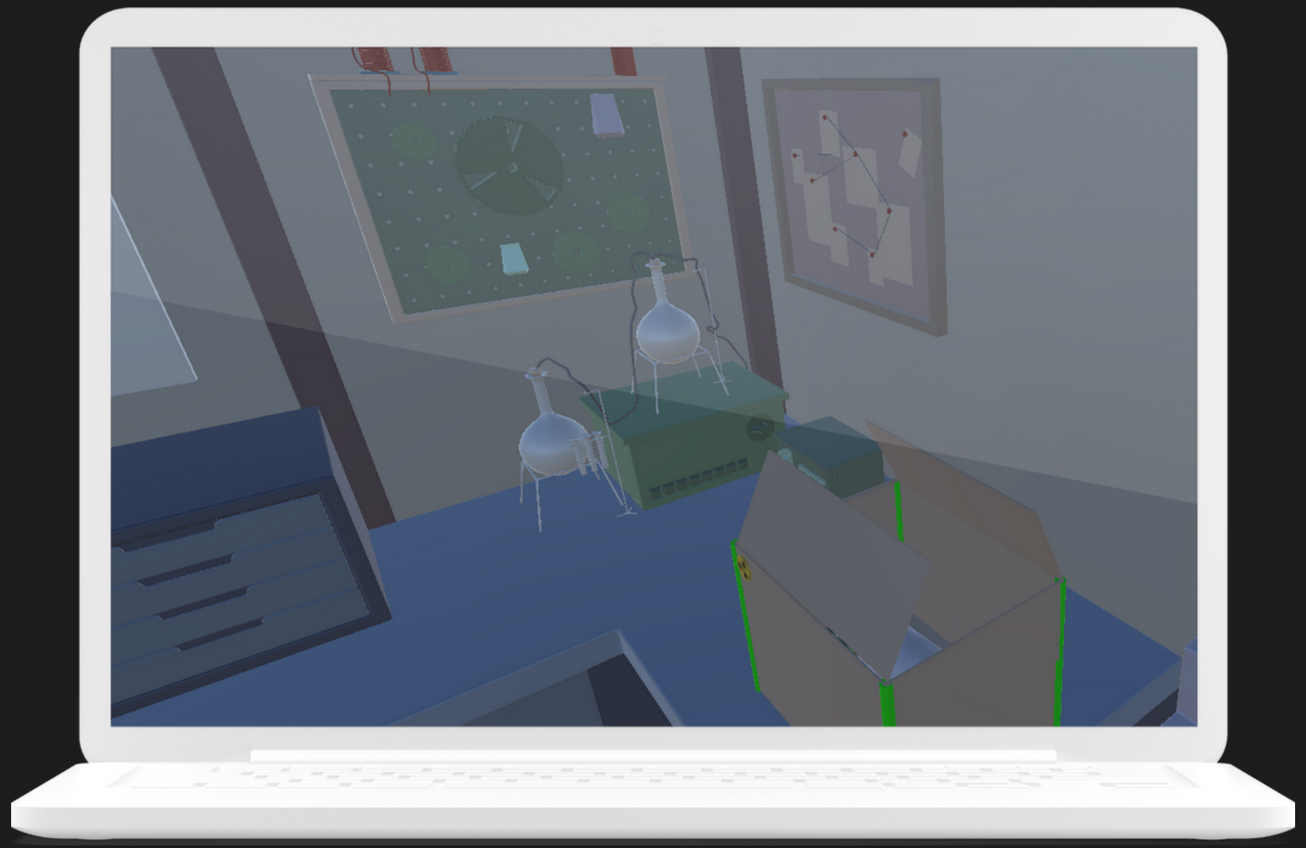
4.Schrodingers Cat

Problem:
Our objective was to delve into the intricacies of the quantum world, aiming to comprehend its underlying phenomena. To achieve this, our focus was on developing a simulation that could effectively visualize the intricate reality of quantum mechanics. The ultimate goal was to create a tool that not only allowed us to explore and understand these complex quantum processes but also to make this knowledge accessible to a broader audience.



Schrodingers Cat

Approach:
We developed an interface for the Schrodinger's cat simulation. Users could interact with the simulation and try to guess whether the cat was alive or not. To make the simulation more accurate, we utilized real quantum computers through IBM's Qiskit solution. Using quantum circuits, we were able to predict the state of the atom during an entanglement and determine the state of the qubit at any given moment



Schrodingers Cat

My contribution

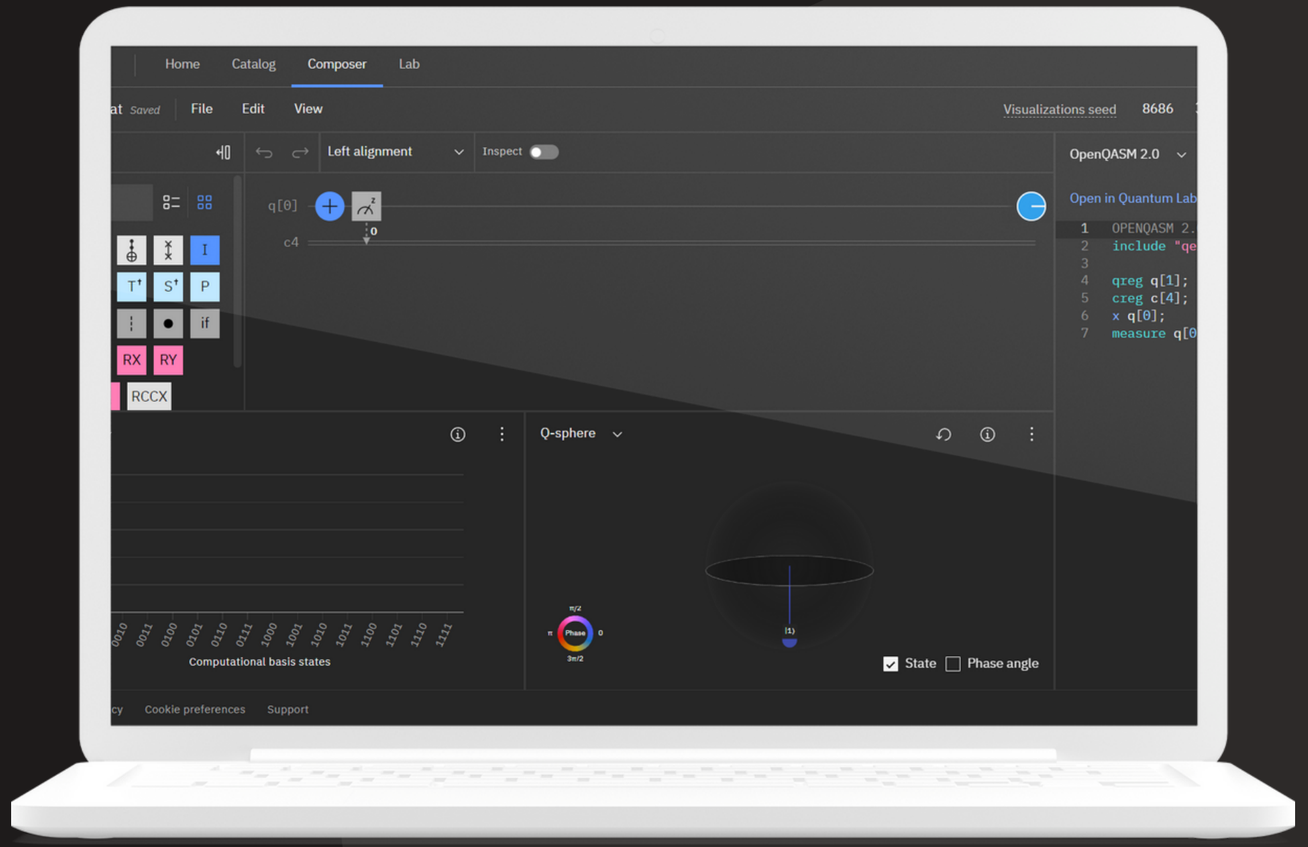
I took charge of the Schrödinger's Cat simulation project, leading the team with a focus on project management, meeting deadlines, and task breakdown. My responsibilities extended to integrating real-life quantum computing using IBM Qiskit, involving practical work with quantum circuits. Beyond managerial roles, I actively contributed to technical aspects, delving into the intricacies of quantum mechanics.



Schrodingers Cat

Impact:

By making this experiment available online, we were able to educate 200 students on the concepts of quantum computing and quantum science. We received 90% positive responses from student, indicating their understanding of these complex concepts. This project had a significant impact on spreading knowledge and fostering interest in the field of quantum science



Schrodingers Cat

Learnings:

One significant learning was the ability to translate intricate quantum concepts into simplified explanations, enhancing my skill in making complex topics accessible to a wider audience. Working on the simulation project also honed my analytical thinking and problem-solving capabilities, as it required tackling challenges inherent in capturing the nuances of quantum phenomena. The experience deepened my knowledge of quantum computing and its applications, providing practical insights into the potential uses and limitations of this groundbreaking technology.

